

Solution:

Q. 1

Since the gate current is zero, the voltage at the gate is simply determined by the voltage divider formed by the two 10-M Ω resistors,

$$V_G = V_{DD} \frac{R_{G2}}{R_{G2} + R_{G1}} = 10 \times \frac{10}{10 + 10} = +5 \text{ V}$$

With this positive voltage at the gate, the NMOS transistor will be turned on. We do not know, however, whether the transistor will be operating in the saturation region or in the triode region. We shall assume saturation-region operation, solve the problem, and then check the validity of our assumption. Obviously, if our assumption turns out not to be valid, we will have to solve the problem again for triode-region operation.

Refer to Fig. 4.23(b). Since the voltage at the gate is 5 V and the voltage at the source is $I_D (\text{mA}) \times 6 (\text{k}\Omega) = 6I_D$, we have

$$V_{GS} = 5 - 6I_D$$

Thus I_D is given by

$$\begin{aligned} I_D &= \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 \\ &= \frac{1}{2} \times 1 \times (5 - 6I_D - 1)^2 \end{aligned}$$

which results in the following quadratic equation in I_D :

$$18I_D^2 - 25I_D + 8 = 0$$

This equation yields two values for I_D : 0.89 mA and 0.5 mA. The first value results in a source voltage of $6 \times 0.89 = 5.34$, which is greater than the gate voltage and does not make physical sense as it would imply that the NMOS transistor is cut off. Thus,

$$\begin{aligned} I_D &= 0.5 \text{ mA} \\ V_S &= 0.5 \times 6 = +3 \text{ V} \\ V_{GS} &= 5 - 3 = 2 \text{ V} \\ V_D &= 10 - 6 \times 0.5 = +7 \text{ V} \end{aligned}$$

Since $V_D > V_G - V_t$, the transistor is operating in saturation, as initially assumed.

Q. 2

Small-Signal Properties

$\lambda = 0$

$V_{in} = -V_{gs}$

$\frac{V_{out}}{R_D} - g_m V_{in} = 0$

$\Rightarrow A_v = g_m R_D$

Rin:

$V_x = -V_{gs}$

$i_x - g_m V_x = 0$

$R_{in} = \frac{V_x}{i_x} = \frac{1}{g_m}$

Application

LNA

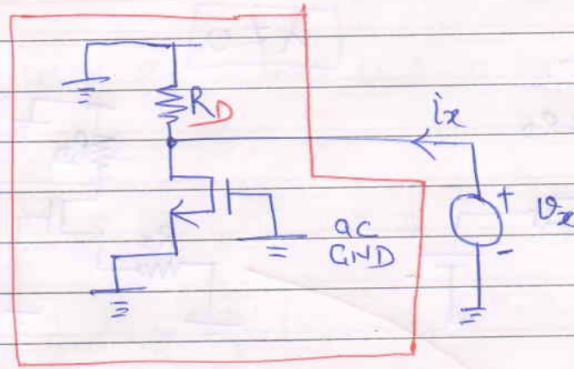
R_{out}

R_{in}

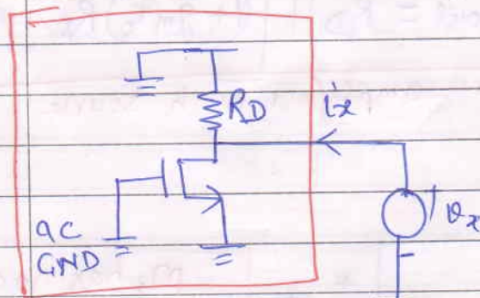
50 Ω

V_{out}

Route:



$$R_{out} = R_D$$

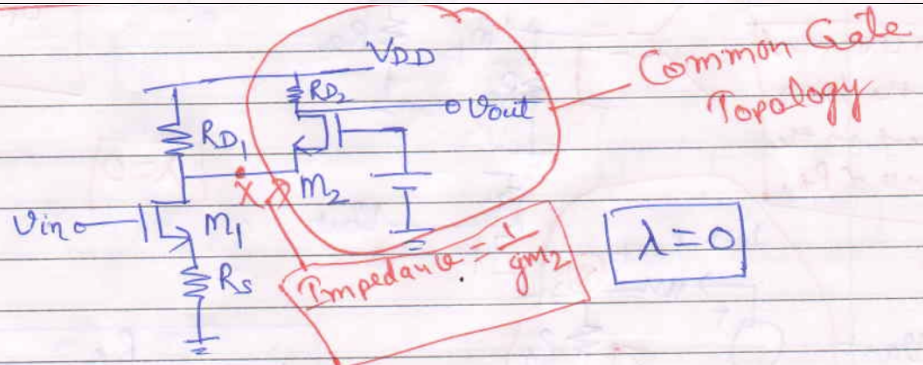


Output resistance of Common Source gate topology is similar to the output resistance of common Source topology.

Common Source Amplifier

Q.3

Example:



Common Gate Topology

$$\lambda = 0$$

$$\text{Impedance} = \frac{1}{g_{m2}}$$

$$\frac{V_{out}}{V_{in}} = \frac{V_x}{V_{in}} \times \frac{V_{out}}{V_x}$$

$$\frac{V_{out}}{V_{in}} = \frac{-\left(R_{D1} \parallel \frac{1}{g_{m2}}\right)}{\frac{1}{g_{m1}} + R_s} \times g_{m2} R_{D2}$$

Q.4

(a) Noting that the source of M_1 is at ac ground, we write the gain as

$$A_v = -g_{m1}[r_{O1} \parallel r_{O2} \parallel (1/g_{m2})]$$

Since $V_{out} = V_{DD} - |V_{GS2}|$, the maximum allowable dc level of V_{in} is equal to $V_{DD} - |V_{GS2}| + V_{TH1}$.

Q.5

(b) If $V_{in} = V_{DD}$, then $V_x = V_{DD} - V_{GS3}$. For M_1 to be saturated when $V_{in} = V_{DD}$, we must have $V_{DD} - V_{GS3} - V_{TH1} \leq V_{DD} - |V_{GS2}|$, and hence $V_{GS3} + V_{TH1} \geq |V_{GS2}|$.